

IMAT MOCK EXAM 10: Final Simulation

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. "Stop taking this medicine and tell your doctor immediately if you notice any of the following rare side effects: swelling of the face, lips or throat, difficulty breathing, or a severe skin rash with blistering. Common side effects include nausea and headache. Uncommon side effects include dizziness and sleep disturbance. The frequency of certain other effects cannot be determined based on available data."

From the informational leaflet for a prescription antihistamine

Which of the following statements is the correct interpretation of the information contained in this excerpt?

- A) Sleep disturbance is a reason to stop the medicine immediately.
- B) Nausea is a rarer side effect of this medicine than a severe skin rash.
- C) A patient who develops a blistering skin rash while taking this medicine should stop taking it and contact a doctor immediately.
- D) Difficulty breathing is a common side effect of this medicine.
- E) Every patient who takes this medicine experiences at least one side effect.

2. "Street trees and small parks measurably cool dense neighbourhoods in summer. In a survey of ten Mediterranean cities, streets whose canopy covered more than 30% of the view from the pavement were on average 2.1 °C cooler on summer afternoons than otherwise similar streets with sparse canopy. The difference was largest in districts with low buildings and narrow streets, where shaded surfaces trap less heat. Cooling depended on species and on irrigation: unwatered young trees grew too slowly in dry summers to shade anything. City planners quoted in the report treat canopy as one instrument among several, alongside reflective roofs and lighter pavements."

From a newspaper report on urban climate research

Indicate which of the following statements is not correct:

- A) The report presents canopy as one of several instruments available to city planners.
- B) How well trees cool a street can depend on the species and on whether the trees are watered.
- C) Streets with more than 30% canopy cover were, on average, cooler than similar streets with sparse canopy in the survey.
- D) Tree planting has been shown by the report to be a complete solution to summer heat in Mediterranean cities.
- E) In districts with low buildings and narrow streets, the cooling difference associated with canopy was largest.

3. "In this country the ruler causes money to be made from the inner bark of mulberry trees, pressed into sheets like paper and cut into round pieces of different values. With this money everything can be bought throughout the whole empire. Refusing it is punishable, and every year the ruler causes all the mulberry bark that can be gathered to be bought up. It costs him almost nothing, yet with it he buys the goods of merchants from every land, who must accept it in exchange."

Adapted from The Travels of Marco Polo (13th century)

What does the passage describe?

- A) Gold coins stamped with the ruler's seal
- B) Barter between merchants and the ruler
- C) Letters of credit drawn on banks
- D) Seashells used as currency
- E) Paper money issued by a ruler

4. "Bacteria in the large intestine ferment fibres that human digestive enzymes cannot break down. The fermentation releases short-chain fatty acids, which are absorbed and used by the cells of the intestinal wall as an energy source. Some of these acids also enter the bloodstream and influence the activity of immune cells. The composition of the bacterial population differs between people and shifts with diet over weeks."

From a popular science book on the human body

Only one of the following statements is coherent with the excerpt:

- A)** Gut bacteria can convert dietary fibre into substances that the body absorbs and uses.
- B)** Immune cell activity is independent of the products of gut bacteria.
- C)** Short-chain fatty acids act only inside the intestinal lumen and never enter the blood.
- D)** Human digestive enzymes release short-chain fatty acids from fibre in the small intestine.
- E)** The bacterial population of the intestine is fixed at birth and does not change with diet.

5. Consider this statement: Every chemist working in this laboratory speaks German.

Which of the following statements is deducible from the given statement?

- A)** Some chemists working in the laboratory do not speak German.
- B)** Someone working in the laboratory who does not speak German is not one of its chemists.
- C)** Nobody in the laboratory speaks German.
- D)** Everyone in the laboratory who speaks German is a chemist.
- E)** German is spoken in the laboratory only by chemists.

6. From the given statements:

If the vaccine shipment arrives on Monday, then the clinic starts the vaccination campaign on Tuesday.

The campaign did not start on Tuesday.

Which conclusion can be deduced?

- A)** The campaign started on Monday.
- B)** The campaign will start on Wednesday.
- C)** The vaccine shipment was lost.
- D)** The vaccine shipment did not arrive on Monday.
- E)** The vaccine shipment arrived on Monday.

7. Five runners, Ada, Beppe, Chiara, Dario and Elena, finished a race with no tied places. Dario finished before Chiara, Chiara finished before Ada, Ada finished before Beppe, and Beppe finished before Elena. Which of the following statements can be deduced?

- A)** Chiara finished last.
- B)** Ada finished before Dario.
- C)** Elena finished after Chiara.
- D)** Dario finished after Ada.
- E)** Beppe finished before Ada.

8. Three friends, Nadia, Omar and Paolo, each drove home in a different car, one red, one blue and one white. Nadia drove neither the red nor the blue car, and Omar did not drive the blue car. Which of the following statements is true?

- A)** Omar drove the red car.
- B)** Nadia drove the blue car.
- C)** Paolo drove the red car.
- D)** Omar drove the white car.
- E)** Paolo drove the white car.

9. A watch gains 3 minutes every hour. It is set to the correct time at 8:00 am. When the true time is 10:00 am, the watch shows:

- A)** 9:57 am **B)** 10:00 am **C)** 10:03 am **D)** 10:06 am **E)** 10:09 am

Section B - Biology (Q10-32)

10. The immediate benefit that lactic fermentation provides to a muscle cell working without enough oxygen is that it:

- A)** produces a large amount of ATP by oxidising lactate completely.
- B)** transports pyruvate into the mitochondrial matrix.
- C)** converts lactate into glycogen for storage.
- D)** releases the energy of reduced FAD for ATP synthesis.
- E)** restores NAD^+ so that glycolysis can continue producing ATP.

11. Which of the following statements about glycolysis is correct?

- A)** It requires molecular oxygen in its first step.
- B)** It takes place in the cytosol and produces a net gain of 2 ATP per glucose molecule.
- C)** It reduces FAD to FADH_2 during the oxidation of glucose 6-phosphate.
- D)** It takes place inside the mitochondrial matrix and produces a net gain of 4 ATP per glucose molecule.
- E)** It converts glucose directly into acetyl-CoA without forming pyruvate.

12. In a eukaryotic cell supplied with oxygen, the largest share of the ATP produced from one glucose molecule comes from:

- A)** glycolysis itself.
- B)** lactic fermentation.
- C)** oxidative phosphorylation.
- D)** the conversion of pyruvate to acetyl-CoA.
- E)** the Krebs cycle operating directly.

13. ATP synthase of the inner mitochondrial membrane produces ATP using the energy released by:

- A)** protons flowing down their gradient through the enzyme.
- B)** diffusion of glucose across the membrane.
- C)** direct transfer of a phosphate group from NADH to ADP.
- D)** hydrolysis of creatine phosphate.
- E)** electrons jumping directly onto oxygen molecules.

14. In the mitochondrial respiratory chain, which carrier transfers electrons from complex I to complex III?

- A)** ATP synthase
- B)** NADH
- C)** Molecular oxygen
- D)** Cytochrome c
- E)** Ubiquinone (coenzyme Q)

15. In a tissue that is metabolising actively and releasing much CO_2 , haemoglobin releases oxygen more readily because:

- A)** the rising pH shifts haemoglobin into a high-affinity state.
- B)** H^+ and CO_2 bind to haemoglobin and lower its affinity for oxygen.
- C)** oxygen bound to one heme group prevents oxygen from binding to the others.
- D)** CO_2 competes with oxygen for the iron atom of the heme group.
- E)** the red blood cells take up bicarbonate and expel all their oxygen.

16. In skeletal muscle, the calcium ions released during contraction bind to troponin. The effect of this binding is to:

- A)** hydrolyse the ATP attached to the myosin heads.
- B)** provide the energy for the power stroke.
- C)** shorten the myosin filaments themselves.
- D)** move tropomyosin away from the myosin-binding sites on actin.
- E)** detach the myosin heads from actin permanently.

- 17.** Cardiac muscle cells are joined end to end by intercalated discs containing gap junctions. The main function of these junctions is to:
- A)** allow rapid electrical coupling between adjacent cells.
 - B)** store oxygen within the muscle cells.
 - C)** increase the surface area available for gas exchange.
 - D)** synthesise the collagen fibres of the heart skeleton.
 - E)** anchor the cells firmly to the heart's tendons.
- 18.** During contraction of a sarcomere, which region remains unchanged in length?
- A)** The I band
 - B)** The I band together with the H zone
 - C)** The A band
 - D)** The H zone
 - E)** None: every band shortens
- 19.** The effect of a competitive inhibitor on an enzyme-catalysed reaction can be overcome effectively by:
- A)** raising the pH to the enzyme's optimum value.
 - B)** cooling the reaction mixture.
 - C)** increasing the concentration of a cofactor of the enzyme.
 - D)** adding a non-competitive inhibitor.
 - E)** increasing the concentration of the substrate.
- 20.** A small organic non-protein molecule that must be bound to an enzyme for the enzyme to show its activity is called:
- A)** a holoenzyme.
 - B)** a coenzyme.
 - C)** a substrate analogue.
 - D)** a zymogen.
 - E)** an apoenzyme.
- 21.** Which digestive enzyme acts in the stomach and has an optimum pH of about 2?
- A)** Pepsin
 - B)** Pancreatic lipase
 - C)** Trypsin
 - D)** Intestinal maltase
 - E)** Pancreatic amylase
- 22.** In the liver, the toxic ammonia produced by the breakdown of amino acids is converted mainly into:
- A)** uric acid.
 - B)** creatinine.
 - C)** glucose.
 - D)** carbon dioxide.
 - E)** urea.
- 23.** If the secretion of antidiuretic hormone (ADH) stops, the most direct consequence for the kidneys is that the person produces:
- A)** urine containing glucose.
 - B)** a small volume of highly concentrated urine.
 - C)** urine with an abnormally high sodium concentration and no other change.
 - D)** a large volume of dilute urine.
 - E)** no urine at all.
- 24.** The cells that resorb bone matrix by secreting acid and digestive enzymes are:
- A)** osteoblasts. **B)** osteoclasts. **C)** chondroblasts. **D)** osteocytes. **E)** fibroblasts.

- 25.** Which hormone raises the concentration of glucose in the blood by stimulating glycogen breakdown in the liver?
A) Melatonin **B)** Insulin **C)** Glucagon **D)** Aldosterone **E)** Thyroxine
- 26.** The bond that joins two glucose units within glycogen is:
A) a peptide bond.
B) a phosphodiester bond.
C) a glycosidic bond.
D) a hydrogen bond.
E) an ester bond.
- 27.** A cell is treated with ouabain, a poison that blocks the sodium-potassium pump. The most direct consequence is that the cell gradually:
A) loses the Na^+/K^+ concentration gradient across its membrane.
B) reverses the direction of the Na^+/K^+ gradient across its membrane.
C) increases its rate of ATP production.
D) starts pumping potassium out and sodium in.
E) becomes freely permeable to all ions.
- 28.** A digestive enzyme secreted by a pancreatic cell reaches the plasma membrane by passing through which sequence of structures?
A) Golgi apparatus, rough ER, lysosome, plasma membrane.
B) Smooth ER, Golgi apparatus, nucleus, plasma membrane.
C) Free ribosome, then nucleus, then plasma membrane.
D) Rough ER, Golgi apparatus, secretory vesicle, plasma membrane.
E) Rough ER, lysosome, Golgi apparatus, plasma membrane.
- 29.** Lysosomal enzymes are synthesised on the rough ER and are then sorted, tagged and packaged for delivery to the lysosomes by:
A) the nucleolus.
B) the Golgi apparatus.
C) the peroxisomes.
D) the smooth endoplasmic reticulum.
E) the mitochondria.
- 30.** A triglyceride molecule consists of:
A) three amino acids joined to glycerol.
B) one glucose unit joined to three fatty acids.
C) glycerol and four fatty acids.
D) glycerol, two fatty acids and one phosphate group.
E) glycerol and three fatty acids.
- 31.** The ability to taste a certain bitter compound is controlled by a dominant allele T. A man who is a heterozygous taster (Tt) and a woman who cannot taste the compound (tt) have a child. The probability that the child cannot taste the compound is:
A) 0 **B)** 1/4 **C)** 1/2 **D)** 1 **E)** 3/4
- 32.** During quiet inhalation, which of the following occurs?
A) The diaphragm contracts and the thoracic volume increases.
B) The diaphragm relaxes and rises back into the thorax.
C) The internal intercostal muscles contract strongly.
D) The pressure inside the alveoli rises above atmospheric pressure.
E) The rib cage is pulled inward.

Section C - Chemistry (Q33-47)

- 33.** A 0.001 mol/dm^3 solution of a monoprotic weak acid has a pH of 4.0. The acid dissociation constant K_a of this acid is approximately:
- A)** 1.0×10^{-3} **B)** 1.0×10^{-4} **C)** 1.0×10^{-5} **D)** 1.0×10^{-6} **E)** 1.0×10^{-7}
- 34.** An aqueous solution of ammonium chloride, NH_4Cl , the salt of a strong acid and a weak base, is:
- A)** impossible to classify without knowing the concentration.
B) acidic, because NH_4^+ donates a proton to water.
C) basic, because Cl^- is a strong base.
D) neutral, because it is a salt and salts do not affect pH.
E) basic, because NH_4^+ releases hydroxide ions.
- 35.** A solid substance does not conduct electricity at all, but the same substance in the molten state conducts well. The substance is most likely:
- A)** glucose.
B) sulfur.
C) copper.
D) sodium chloride.
E) silicon dioxide.
- 36.** Which of the following bonds is the most polar? (Pauling electronegativities: H = 2.20, C = 2.55, N = 3.04, O = 3.44, Cl = 3.16)
- A)** C-N **B)** C-Cl **C)** C-H **D)** C-C **E)** C-O
- 37.** What is the average oxidation number of sulfur in sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$?
- A)** -2 **B)** 0 **C)** +2 **D)** +4 **E)** +6
- 38.** In the reaction $2\text{Mg} + \text{CO}_2 \rightarrow 2\text{MgO} + \text{C}$, which species is oxidised?
- A)** Magnesium, because its oxidation number rises from 0 to +2.
B) CO_2 , because it loses oxygen atoms.
C) Carbon, because its oxidation number falls from +4 to 0.
D) Oxygen, because it changes partner.
E) Both magnesium and carbon are oxidised.
- 39.** Which of the following compounds is a ketone?
- A)** Propan-1-ol **B)** Methanal **C)** Propanal **D)** Propanoic acid **E)** Propanone
- 40.** Which of the following pairs contains only monosaccharides?
- A)** glucose and sucrose
B) starch and cellulose
C) maltose and fructose
D) glucose and fructose
E) lactose and galactose
- 41.** What amount of oxygen, in moles, is required for the complete combustion of 0.5 mol of propane? ($\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$)
- A)** 0.5 mol **B)** 1.5 mol **C)** 2.0 mol **D)** 2.5 mol **E)** 5.0 mol
- 42.** What mass of water is produced when 4.0 g of methane burns completely in excess oxygen? ($\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$; molar masses: $\text{CH}_4 = 16 \text{ g/mol}$, $\text{H}_2\text{O} = 18 \text{ g/mol}$)
- A)** 4.5 g **B)** 8.0 g **C)** 9.0 g **D)** 18.0 g **E)** 36.0 g

- 43.** Going from left to right across a period of the periodic table, the first ionisation energy of the elements generally:
- A)** decreases, because the outer electron lies further from the nucleus.
 - B)** increases, because nuclear charge rises while the outer electrons enter the same shell.
 - C)** increases, because the atomic radius increases.
 - D)** remains constant, because the shell number does not change.
 - E)** decreases, because the atoms become heavier.
- 44.** The maximum number of electrons that the 3p sublevel can hold is:
- A)** 2 **B)** 3 **C)** 6 **D)** 8 **E)** 10
- 45.** The industrial synthesis of ammonia is exothermic: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, $\Delta H < 0$. Which set of conditions gives the highest equilibrium yield of NH_3 ?
- A)** Low temperature and low pressure
 - B)** Any temperature, provided a catalyst is used
 - C)** High temperature and low pressure
 - D)** High temperature and high pressure
 - E)** Low temperature and high pressure
- 46.** Adding a catalyst to a reversible reaction already at equilibrium:
- A)** shifts the equilibrium position towards the products.
 - B)** lowers the activation energies of forward and reverse reactions and does not shift the position of equilibrium.
 - C)** raises the activation energy of the forward reaction only.
 - D)** increases the concentrations of the reactants.
 - E)** increases the enthalpy change of the reaction.
- 47.** The compound CuSO_4 is correctly named:
- A)** copper(II) sulfate.
 - B)** copper(II) sulfide.
 - C)** copper(I) sulfate.
 - D)** copper(I) sulfide.
 - E)** copper(II) sulfite.

Section D - Mathematics (Q48-54)

- 48.** The set of real solutions of the inequality $x^2 + 2x + 5 \leq 0$ is:
- A)** $x \leq -1$.
 - B)** the set of all real numbers.
 - C)** only $x = -1$.
 - D)** the empty set.
 - E)** $x \geq -1$.
- 49.** Solve the inequality $5x - 3 < 2x + 9$.
- A)** $x > 4$ **B)** $x < 2$ **C)** $x > 2$ **D)** $x < 12$ **E)** $x < 4$
- 50.** The equation $x^2 - 5x + k = 0$ has one root equal to 2. The value of k is:
- A)** -6 **B)** 2 **C)** 4 **D)** 6 **E)** 10
- 51.** If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, how many elements does $A \cup B$ contain?
- A)** 4 **B)** 5 **C)** 6 **D)** 7 **E)** 8
- 52.** The product $(2x - 3)(x + 5)$ is equal to:
- A)** $2x^2 + 7x - 15$ **B)** $2x^2 - 7x - 15$ **C)** $x^2 + 7x - 15$ **D)** $2x^2 + 13x - 15$ **E)** $2x^2 + 7x + 15$
- 53.** A rectangle's length is 3 cm greater than its width, and its perimeter is 26 cm. Its width is:
- A)** 4 cm **B)** 5 cm **C)** 6 cm **D)** 8 cm **E)** 10 cm

54. If $\sin \theta = 0.6$ and θ is an acute angle, then $\cos \theta$ is:

- A)** 0.36 **B)** 0.4 **C)** 0.6 **D)** 0.8 **E)** 1.0

Section E - Physics (Q55-60)

55. A car travels along a straight motorway at a constant 90 km/h, following a truck that travels at a constant 72 km/h and is 2.0 km ahead. After 5 minutes, which statement is correct?

- A)** The car needs more than a further hour to catch the truck.
B) The car has overtaken the truck.
C) The car has drawn exactly level with the truck.
D) The car will never catch the truck.
E) The car has not yet caught the truck.

56. A circuit contains two resistors of $6 \, \Omega$ each connected in parallel, and this pair is connected in series with a single $3 \, \Omega$ resistor. The equivalent resistance of the whole network is:

- A)** $2 \, \Omega$ **B)** $3 \, \Omega$ **C)** $6 \, \Omega$ **D)** $9 \, \Omega$ **E)** $15 \, \Omega$

57. A wire of uniform material is replaced by a new wire that is twice as long and has half the cross-sectional area. Compared with the original wire, the resistance of the new wire is:

- A)** unchanged.
B) four times greater.
C) half.
D) four times smaller.
E) double.

58. At what height above the Earth's surface is the acceleration due to gravity equal to one quarter of its value at the surface? (The Earth's radius is R .)

- A)** R **B)** $2R$ **C)** $R/4$ **D)** $R/2$ **E)** $4R$

59. A car of mass $1.5 \times 10^3 \, \text{kg}$ accelerates along a level road at $2.0 \, \text{m/s}^2$ against a constant frictional force of 500 N. The force the engine must exert is:

- A)** 500 N **B)** 2500 N **C)** 3000 N **D)** 3500 N **E)** 4000 N

60. A swimming pool is 4 m deep. Taking the density of water as $1000 \, \text{kg/m}^3$ and $g = 10 \, \text{m/s}^2$, the gauge pressure of the water at the bottom of the pool is:

- A)** 400 Pa **B)** 2500 Pa **C)** 4000 Pa **D)** 40000 Pa **E)** 400000 Pa

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	C	2	D	3	E	4	A	5	B
6	D	7	C	8	A	9	D	10	E
11	B	12	C	13	A	14	E	15	B
16	D	17	A	18	C	19	E	20	B
21	A	22	E	23	D	24	B	25	C
26	C	27	A	28	D	29	B	30	E
31	C	32	A	33	C	34	B	35	D
36	E	37	C	38	A	39	E	40	D
41	D	42	C	43	B	44	C	45	E
46	B	47	A	48	D	49	E	50	D
51	C	52	A	53	B	54	D	55	E
56	C	57	B	58	A	59	D	60	D